Definitions

- **Type I Error (False Positive):** Rejecting a true H_0; probability = α.
- **Type II Error (False Negative):** Failing to reject a false H_0; probability = β.

Outcomes Summary

- H_0 true & reject H_0 → **Type I error** (probability α).
- H_0 true & fail to reject → **Correct** (confidence = $1 - \alpha$).
- H_0 false & reject H_0 → **Correct** (power = $1 - \beta$).
- H_0 false & fail to reject → **Type II error**.

Contextual Examples

- **Two-sample t-test:** Type I → conclude two groups differ when they do not; Type II → miss a real difference.
- **Chi-square test of independence:** Type I → claim association when variables are independent; Type II → miss a true association.

Managing Errors

- Lower α reduces Type I risk but usually increases β; higher α does the opposite.
- Increasing sample size reduces both errors by decreasing standard error.
- Choosing more sensitive tests or better measurement instruments can increase power ($1 - \beta$).

Communication Tips

Always describe errors in context: specify the actual claim (means differ, proportions equal, etc.) so students grasp consequences. Avoid saying "accept H_0"; instead use "fail to reject H_0."

Related Documents

- `test_power_and_examples.pdf` explains how to quantify β and power.
- Per-topic how-to PDFs provide scenario-specific Type I/II statements.